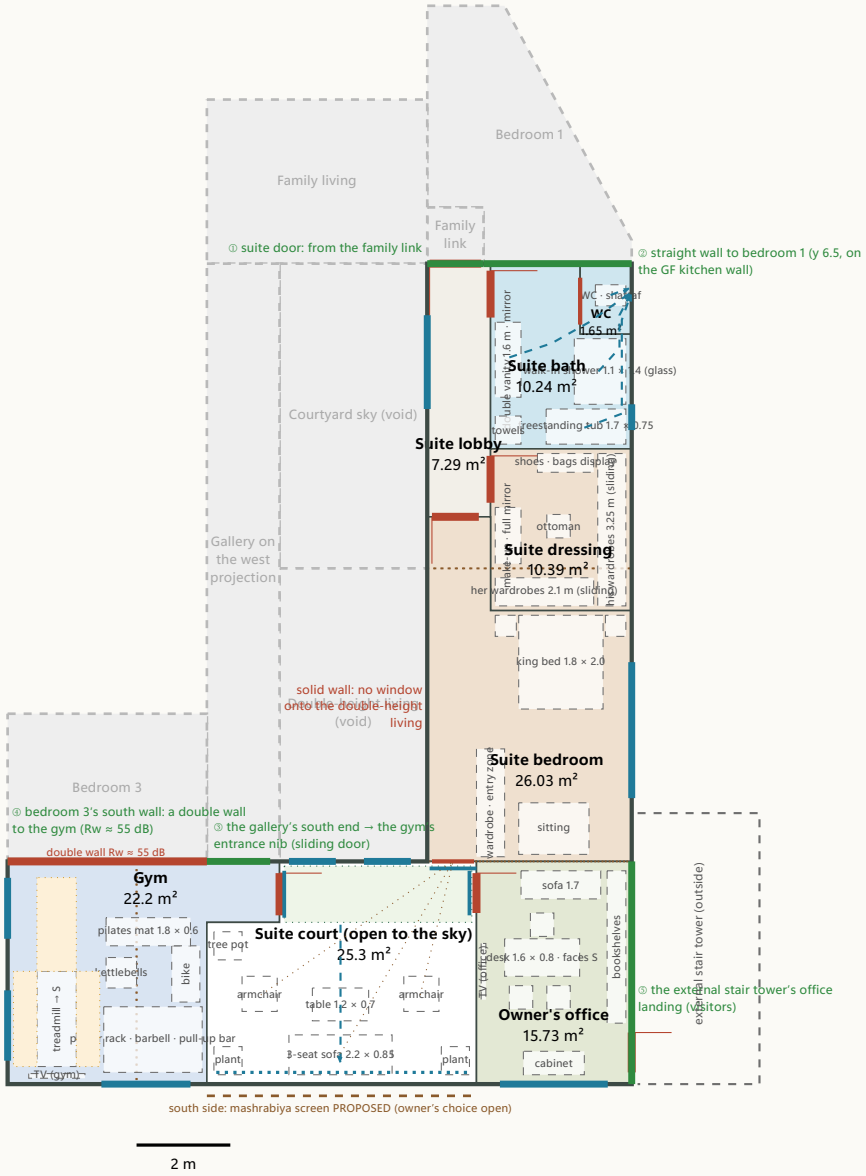
First floor · the south wing v2 (gym west, the suite court, the office east) and the master suite

Taif / Palm House · CSV

coordinates (x east, y south) · wall centre lines · north up

First floor, part 2 of 2 (part 1: studies/ff_family.typ) · owner's decision 1 Nov 2026 (south wing v2): the gym on the west band, the suite court from the gym to the office, the office on the east, the three court doors on a 1.2 m strip; the master suite unchanged



Rooms

Room	Axes w × d (m)	Axes m²	≈ clear m²
Suite lobby	1.35 × 5.4	7.29	5.94
Suite bath	polygon	10.24	8.79
Suite WC (closet)	1.1 × 1.5	1.65	1.09
Suite dressing	3.01 × 3.45	10.39	9.01
Suite bedroom	polygon	26.03	23.56
Gym	polygon	22.2	19.87
Suite court (open to the sky)	polygon	25.3	23.1
Owner's office	3.31 × 4.75	15.73	13.91
Zone		**118.83**	

Clear ≈ axes minus half a wall on every edge (250 mm outer, 200 mm inner), corners ignored; the court: 5.55 × 4.525 inside its walls minus the nib.

✓ All rooms inside the zone · no overlaps · no unassigned area · furniture inside its room, not overlapping, out of the routes · door swings clear and not clashing · the three court doors on the 1.2 m strip · the court seating out of the strip · the treadmill with its manufacturer zone (0.5 m each side, 2 m behind) · the bed out of sight from the gym door, the office door and the court seating (doors open) · no door bedroom ↔ office

Key numbers

- **Suite:** lobby 7.29 · bath 10.24 + WC closet 1.65 · dressing 10.39 · bedroom 26.03 = **55.6 m² indoors** + the suite court.
- **Gym** 22.2 m² on axes (≈ 19.87 clear; the west band alone 20.19 / 18.21) · **suite court** 25.3 m² on axes (≈ 23.1 clear) · **owner's office** 15.73 m² (≈ 13.91 clear).
- **Walked:** gallery end → gym → court → office door ≈ 5.8 m; office → the stair landing ≈ 4.8 m; bed → court ≈ 4.8 m.

Sight lines into the suite bedroom (its court door open; plan)

From	without the fix	with the entry wardrobe
the gym's court door (open), standing in it	bed hidden	bed hidden
the office's court door (open), standing in it	bed hidden	bed hidden
the court seating: the sofa	the bed is seen	bed hidden
the court seating: the west armchair	bed hidden	bed hidden
the court seating: the east armchair	the bed is seen	bed hidden
the court seating: the table	the bed is seen	bed hidden

Rays through the 0.9 m door opening (0.2 m wall) into the room, stopped by the walls and the wardrobe; the bed 1.8 × 2.0 at x 10.9–12.7, y 14–16.

Plan: red = doors (thin = sliding) with swing line · blue = windows, blue lines = the court's linear drains, dotted = its box gutter · green dotted = the 1.2 m movement strip · brown dotted = ground floor below and the sight lines · grey = the family part (fixed).

Master suite, suite court, gym and office · rooms, the court's details, walls, questions

Suite lobby, bath, dressing (unchanged)

- The suite door (from the family link) opens into a distributor: bath, dressing and bedroom each open off it; the bedroom door at the lobby's south end (a bent entry). Luxury bath 11.89 m² with tub, walk-in shower, double vanity and a WC closet; walk-in dressing 3.01 × 3.45 m.

Suite bedroom · 26.03 m² · privacy without a screen

- **King bed** on the dressing wall facing south; the sitting corner by the east (escape) window; **no door to the office** (owner).
- **The court door** (0.9 m sliding glass, x 9.05–9.95) opens onto an **entry zone** inside the room (x 8.95–10.0, y 16.85–19.25) formed by a full-height **wardrobe 0.6 × 2.3** east of it: the **smallest fix** the sight-line test needs. Without it the bed is seen from the court seating through the open door (table on page 1); with it the bed is hidden from the gym door, the office door and every seat. The bedroom has no window on the court: the glass door gets a curtain.

Suite court · 25.3 m² on axes, ≈ 23.1 m² clear · open to the sky, four sides

- From the gym's east wall (x 4.25) to the office (x 10), y 19.25–24.0: 5.75 × 4.75 on axes less the gym's entrance nib (1.55 × 1.3) in its north-west corner; 5.55 × 4.525 inside its walls. Sides: north the double-height living's south wall with its clerestory windows (+5.00 … +6.00, 1.40 m above the court floor) and the bedroom's court door; west the gym (solid, its court door on the nib); east the office (solid, its court door); south the facade frame (the cornice and the parapet run across on R8; the infill is the owner's choice: a solid wall, a mashrabiya screen (recommended, **proposed**) or a 1.1 m wall, `studies/south_wing.typ` § v2).
- **Movement strip:** the three court doors (bedroom: sliding; gym: hinged, into the court; office: hinged, into the office) open onto a **1.2 m clear strip** along the north edge (x 5.9–9.9, y 19.35–20.55); the swings do not clash (checked). The seating zone is in the south part, out of the strip.
- **Furniture test:** a 3-seat sofa against the south side, two armchairs and a table 1.2 × 0.7: seating for five; plants in the corners and a tree pot by the gym wall; a free band ≥ 1.0 m from the strip to the seats (checked).
- **Near-flush doors (note for the engineer):** the court floor falls **south** at 1.5 % away from the doors to a box gutter along the south side (two outlets, internal downpipes in the south wall) with an **overflow spout** through the south wall (x ≈ 7.1, below the threshold level); **linear slot drains** in front of each of the three doors; the membrane turned up ≥ 150 mm behind the slot drains and under the door sills (a low-threshold detail: threshold ≤ 20 mm, the drain grating flush); the court's structural slab set 0.10–0.15 below the first-floor slab (a depressed zone in the men's majlis roof, its soffit unchanged) so the falls, the screed and the tiles end flush with the indoor floors.

Gym · the west band 5.8 × 4.75 (axes, incl. the nib), 22.2 m², ≈ 19.87 m² clear

- **v2 (owner):** the gym on the west band only (x 0–4.25, 4.25 × 4.75 on axes, 4.03 × 4.52 clear) plus a small **entrance nib** at the gallery end (x 4.25–5.8, y 19.25–20.55): the gallery and the gym only touch at a corner, so the nib takes the family's sliding door from the gallery end and the gym's door to the court. No south arm. Windows west (two) and south (x 2.0–3.9); one fixed TV on the south wall in front of the treadmill.
- **Treadmill** (Life Fitness T5 class (assumed): 2.03 × 0.84 m) along the 4.75 m length, facing south, 0.5 m clear each side and **2 m free behind it**, 0.5 m off the west wall face: on the **2.75 m solid bay between the west outer wall and beam B10** (over the guest block's walls: the stiffest bay of the wing). The rest: a power rack with the barbell and the pull-up bar, kettlebells, the bike and the pilates mat on the east half; a 1.0 m walkway from the gallery door (checked).
- **What carries it (traced, `studies/structure.typ`):** the gym sits inside the column frame C6 / C13 / C19 (y 19.25) – C7 / C14 / C20 (y 24): the west outer wall on the ground-floor west wall; B10 on x 2.75 (C13 – C14, over the foyer's 1.8 m opening); **B19 on x 4.25 between C19 and C20** (the gym's east wall); the double wall on y 19.25 on B8 (x 2.75–4.25) and on a new beam B22 (C6 – C13, over the guest passage and the WC door); the south wall on B9.
- **Acoustics:** the **double wall to bedroom 3** (y 19.25, x 0–4.25): 200 mm dense block + an independent stud lining with mineral wool, two layers of acoustic board, target **Rw ≈ 55 dB** (≈ 50 dB on site), sealed at the top, no sockets back to back; nothing fixed to it (the rack is freestanding).

Owner's office · 3.31 × 4.75 m on axes, 15.73 m² (≈ 13.91 m² clear)

- East, beside the suite bedroom, **no door between them**. The desk faces the south window (x 10.5–12.8); a sofa on the suite wall; bookshelves on the east wall; one fixed TV on the court wall. **Doors:** the visitors' door from the external stair tower's landing (east wall, y 22.9–23.75, opening out onto the landing, thumb-turn inside) and the court door (x 10, y 19.5–20.35, opening into the office: **free from the court side, key only from the office side**, so visitors cannot enter the court or the house).

Locks on the route house → court → office → stair → outside

Door	opens	lock
court (sliding glass)	sliding	free from the bedroom side
court (from the gym)	into Suite court (open to the sky)	free from the gym side, no key
office (from the court)	into Owner's office	free from the court side; key only from the office side
visitors' door (stair landing)	out of its room	thumb-turn inside; key outside
stair foot gate	outwards	panic bar; key / code outside

Questions for the owner

1. The court's south side: solid wall, mashrabiya screen (recommended) or a 1.1 m wall (`studies/south_wing.typ`).
2. The clerestory windows on the court: fixed fire-resisting glazing if the court is to count as an escape route (`studies/fire_escape.typ`), or openable vents as today.